Noninvasive Estimation of Motor Unit Conduction Velocity Distribution Using Linear Electrode Arrays

Dario Farina, Elena Fortunato, and Roberto Merletti*

***Abstract*—Determining the conduction velocity of motor unit action potentials is one of the most important problems in surface electromyography. The estimate of one average conduction velocity value depends on a variety of uncontrollable factors. More meaningful information is obtained from the estimation of the distribution of the different delays in the myoelectric signals. A solution to the problem is the separation and characterization of the individual components propagating at different velocities. A technique, based on surface electrode array recording, is proposed to estimate motor unit conduction velocity distribution. The method consists in the identification of the single action potentials in the time scale domain (with the continuous wavelet transform) and in the estimation of their conduction velocities based on the beamforming algorithm. The performances of the technique have been evaluated using simulated and real myoelectric signals. The results demonstrate that the technique is accurate and reliable. The method may be useful for the diagnosis of neuromuscular disorders, for the monitoring of muscle fatigue and for noninvasive investigation of individual motor units.**

***Index Terms*—Beamforming, conduction velocity estimation, continuous wavelet transform, electrode arrays, electromyography, spatial filters.**

I. Introduction

MUSCLE fiber conduction velocity (CV) is a basic physiological parameter and it is known to be related to the type and diameter of muscle fibers, ion concentration, pH and motor unit firing rate [1]–[4]. Several methods have been proposed in literature to estimate CV by means of surface electrodes. Most of them are global techniques in the sense that the estimated velocity is an averaged value calculated from a signal composed by a large number of contributions [5]–[10]. A few techniques have been introduced to detect the CV of individual motor units but they are limited by some strong hypotheses and not easy to apply to real cases [11]. In clinical research activity CV distribution of the motor units is obtained only with painful and time consuming invasive techniques [12], [13].

A possible approach to the problem of separating the different contributions to the surface myoelectric (sEMG) signal is the use of linear electrode arrays which provide a spatial sampling of the potential distribution along the muscle during a contraction. The applications of such arrays have been described in the last 15 years and have been mostly focused on the estimation of the location of the neuromuscular junctions [14]–[17]. Since the electrode arrays provide multiple observation points along the entire muscle, the propagation effect is much more evident than in the case of two pairs of detection electrodes and it is possible to recognize and classify the single motor unit action potentials (MUAP's) using information from all channels. It was shown [16] that some MUAP's can be identified, extracted and superimposed, either by visual or computer assisted analysis, to generate clusters. The possibility of recognizing individual motor units (MU's) opens up interesting perspectives in muscle investigation by sEMG signal. These studies represent the starting point of this work, whose purpose is to present a method for the segmentation of the sEMG signal detected with linear electrode arrays and the separation of the contributions travelling at different velocities.

The technique takes advantage from spatial selectivity obtained with spatial filtering [18], [19] and spatial sampling of the sEMG signal. The spatial filtering technique used in this study is the simple double differential technique, but the method can be easily extended to more complex detection systems which permit to improve spatial selectivity and to simplify the separation of the different MUAP's, as shown in literature [20], [21].

II. Method

A. MUAP Detection using Wavelet Transform

The MUAP identification is based on continuous wavelet transform (CWT) and is carried out on a single signal of the array, taken as reference signal. This reference signal is chosen from the array by the operator such that similar time shifted signals are present distally or proximally indicating unidirectional propagation. The first step consists in selecting a group of signals with the same direction of propagation (i.e., detected between the innervation zone and the tendon) and choosing as reference signal the first one of the selected group. For the time being this selection is performed manually. Fig. 1 reports examples of selection of signals from an array of 16 electrodes.

For a time-domain signal $s(t)$, the corresponding CWT is defined as [22]

$$WT(a,\tau) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} s(t) \cdot h^* \left(\frac{t-\tau}{a} \right) dt \qquad (1)$$

where

$h(t)$ is an appropriate function called mother wavelet;

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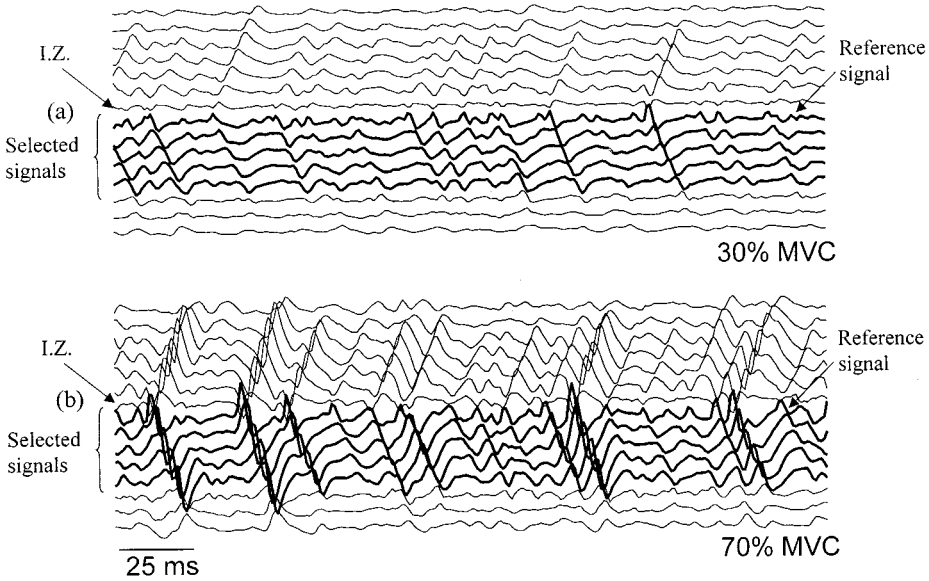

Fig. 1. Signals detected with a linear electrode array of 16 electrodes (interelectrode distance of 10 mm) from the biceps brachii of a healthy subject during (a) 30% maximal voluntary contraction (MVC) and (b) 70% MVC; the selected signals and the reference signal used for the estimation of CV distribution are shown. I.Z.: innervation zone.

- a is the wavelet scale ($a > 0$), reflecting the frequency information of $s(t)$;
- τ is a translation index indicating the waveform shifting in time.

The square of the absolute value of the CWT is called scalogram. From (1) it is clear that the CWT can be interpreted as the result of the application to the signal of filters with impulse responses which are the scaled version of $h^*(-t)$.

The observation that the different MUAP's, detected with surface technique, can be considered as dilated and attenuated versions of one single shape, due to the filtering effect of the volume conductor, leads to the idea of selecting a mother wavelet similar to this shape. The aim is to match the wavelet to the shape of the signal to be detected (as in the case of the matched filter), so the wavelet is chosen according to the detection technique used during EMG recordings. The first-order Hermite–Rodriguez (HR) function (see the Appendix) has been used for example to show the possibility of detecting simulated single differential waveforms [23], while the HR expansion of the M waves is presented in [24].

In this work, the double differential system is used. It presents good spatial selectivity and the second-order HR function matches very well with the detected MUAP's (Fig. 2). Fig. 3(a) shows the scalogram of a simulated double differential signal [Fig. 3(b)] using the second-order HR function as mother wavelet.

According to this technique, in the ideal case of MUAP's which are scaled versions of the mother wavelet, the maximum of the scalogram is achieved at the time instant and scale value in which the corresponding wavelet reproduces exactly the signal. In this case the section of the CWT passing for the scale corresponding to the maximum point is the output of the filter matched to the MUAP; this filter permits to obtain the maximum signal-to-noise ratio (SNR). The maximum point in the time scale domain allows to determine the time location and to

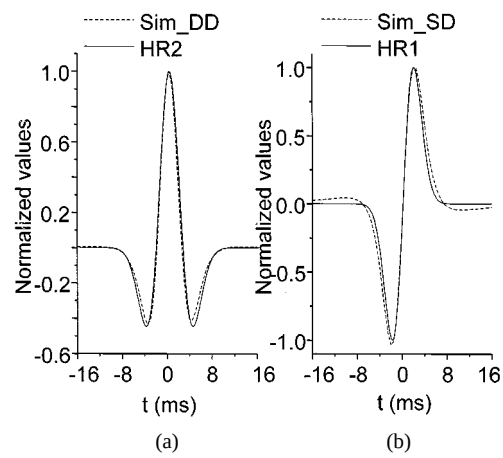

Fig. 2. Matching of (a) the second-order HR function and (b) first-order HR function with a simulated double differential and single differential detected MUAP. The value of λ (5) is the one that minimizes the mean square error.

estimate the time duration of the MUAP. An easy way to detect the maximum point of the scalogram is based on the observation that, if some hypotheses are met, the maximum point τ_M of the scalogram in time is the same for all constant scale sections, as demonstrated in the Appendix. This property permits to detect the time coordinate of the maximum point of the scalogram solving a one-dimensional (1-D) problem instead of a two-dimensional (2-D) one. In fact, considering the time domain marginal distribution of the scalogram $\beta(\tau)$ obtained from its integration in the scale domain, the previous property leads to

$$\beta(\tau) = \int_{-\infty}^{+\infty} |WT(a,\tau)|^2\, da \le \int_{-\infty}^{+\infty} |WT(a,\tau_M)|^2\, da = \beta(\tau_M) \quad \forall \tau. \tag{2}$$

The time location information is completely preserved in the marginal distribution in time while the scale information is

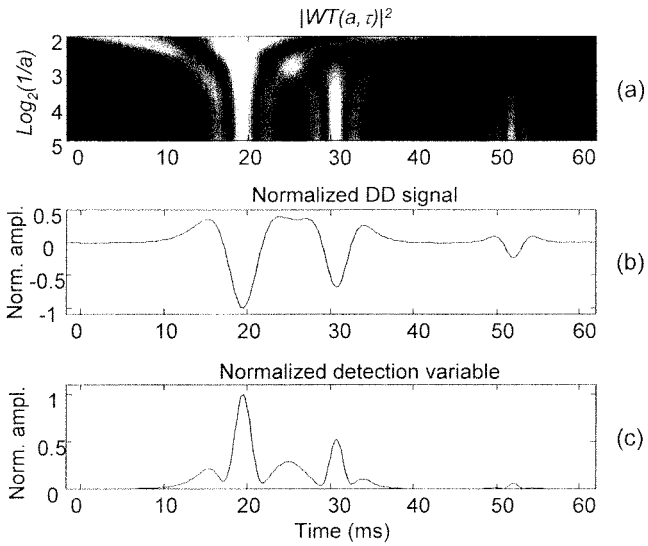

Fig. 3. Scalogram (a) of a simulated double differential signal (b) and its marginal distribution in time domain (detection variable) (c). The mother wavelet is the second-order HR function.

lost. The marginal distribution is used as detection variable [Fig. 3(c)]. The dominant peaks of this variable correspond to MUAP's. During an epoch of myoelectric signal, the detection of a MUAP is acknowledged as corresponding to a local maximum of the signal $\beta(\tau)$.

The time duration of the MUAP is equal to the time duration of the best matched wavelet to the MUAP, which is the wavelet corresponding to the scale s_M where the scalogram, evaluated for $\tau = \tau_M$, is maximum. The time duration of the wavelet is defined as the number of samples of the absolute value of the function greater than 1% of the function maximum. Note that the local maximum point of the 2-D scalogram is found with two searches, each of which finds a maximum point in 1-D space (first time and then scale). The coordinates τ_M and s_M, respectively, provide the location and scale factor (width) of a time-domain window which includes the MUAP or combinations of superimposed MUAP's. The identified MUAP is used as reference waveform in the subsequent array-processing algorithm.

B. Decision Phase and CV Estimation

Beamforming is the name given to a wide variety of array processing algorithms that, by some means, focus the array's signal capturing abilities in a particular direction [25]. Delay and sum beamforming is the basic idea used in this study. If a propagating signal is present in the electrode array and the electrode outputs are delayed by appropriate amounts and added together, the signal is reinforced with respect to either noise or waves propagating at different velocities. The traditional beamforming technique gives an estimation of the delay between signals using a maximum energy deduction criterion: the delay that maximally reinforces the output signal (that is the value which maximizes the energy of the output) coincides with the interval of time the signal takes to propagate between one electrode set and the next. In our case this basic concept is used to estimate the CV and to decide if the detected waveform is a travelling potential or not.

Suppose to have only one action potential which travels at the constant velocity v without shape changes. Consider a linear array of N electrodes, each of which is separated from its neighbors by Δ meters, along the z-axis, parallel to the direction of propagation of muscle fibers. The position of the mth electrode is $m \cdot \Delta$, $m = 0, 1, \cdots, N-1$. The diagram reported in Fig. 4 shows the situation in the spatio-temporal domain. The signal recorded by the mth electrode is $s_0(t - m \cdot \theta_T)$, where $s_0(t)$ is the signal at the first electrode, taken as reference and θ_T is equal to Δ/v.

For each value of θ_i, the $N-1$ signals $s_1(t)$, $s_2(t)$, $\cdots$, $s_{N-1}(t)$ are shifted by θ_i, $2\theta_i$, $\cdots$, $(N-1)\theta_i$, respectively, and added; the output signal is then compared to the reference signal. In ideal conditions the delay-and-sum beamformer's output signal is

$$\begin{aligned} s_{\text{sum}}(t) &= \sum_{m=1}^{N-1} s_m(t + m \cdot \theta_i) \\ &= \sum_{m=1}^{N-1} s_0(t - m \cdot \theta_T + m \cdot \theta_i) \end{aligned} \tag{3}$$

where θ_i is the assumed delay at the ith iteration. The signal $s_{\text{sum}}(t)$ can be interpreted as the output of a spatio-temporal filter (shift in time domain and summation in space domain). If the chosen delay is $\theta_i = \theta_T = \Delta/v$, the delayed signals add constructively. In this special case, the output of the beamforming becomes

$$s_{\text{sum}}(t) = \sum_{m=1}^{N-1} s_0(t - m \cdot \theta_T + m \cdot \theta_i) = (N-1) \cdot s_0(t) \tag{4}$$

which is equal to an undistorted replica of the propagating signal recorded at the first electrode. On the other hand, if a wrong delay is selected, a degraded version of the propagating signal is obtained; in such cases the beamformer is mismatched to the propagating MUAP.

The correlation coefficient (CC), computed for varying θ_i, has been chosen as a similarity measure to evaluate the resemblance between the reference MUAP and the output of the spatio-temporal filter. The optimal delay θ_{op} among all possible delays corresponds to the maximum CC. The CC between the reference MUAP and the output of the beamforming corresponding to θ_{op} is then used to decide if the detected waveform is a MUAP or not. If CC is greater than a preset threshold (0.7–0.8) and if the duration of the identified MUAP is in a preset interval of time, the waveform and the estimated conduction velocity (CV $= \Delta/\theta_{op}$) of the detected MUAP are stored. Note that the CC is maximum where the mean square error (MSE) is minimum. The threshold on the CC permits to distinguish between cases in which a real MUAP has been detected in the first phase (based on the CWT) and cases in which only noise has been detected; in this way the probability of false detection is reduced.

The estimates of delays are implemented entirely in the frequency domain, so that the restrictions on the time resolution are avoided [9]. It should be noted that the $N-1$ electrode signals are shifted in a window whose width is sufficient to include the propagation time of the MUAP to the last electrode. This is also the window used for the calculation of the CWT (referred

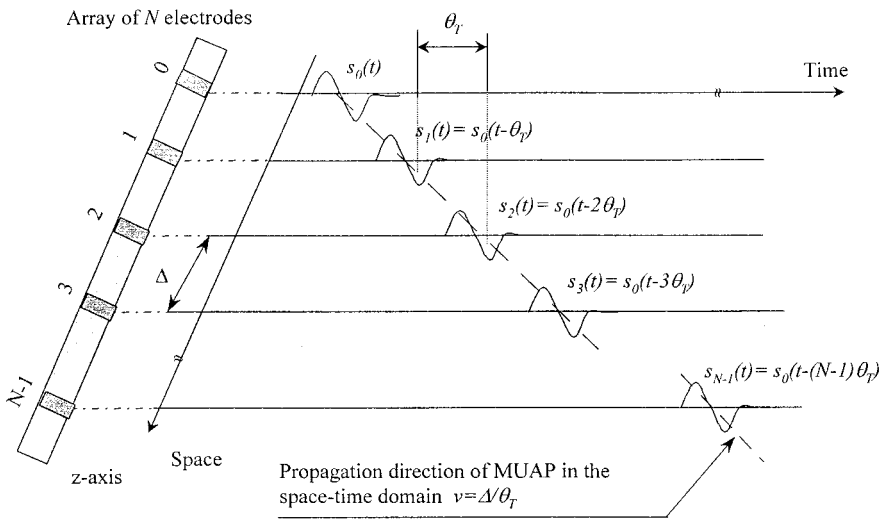

Fig. 4. Propagation of a single MUAP signal along an array of N electrodes. The conduction velocity of the action potential is v and its propagation delay is θ_τ.

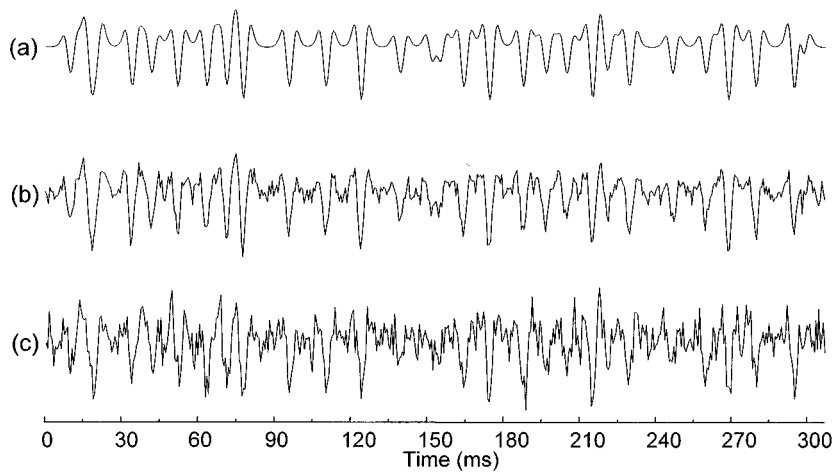

Fig. 5. 300 ms of a 1-s-long simulated double differential signal generated by 92 MUAP's from 11 MU's with some cases of partially overlapping MUAP's (in the simulation seven channels of the array are selected), (a) in the case of infinite SNR, (b) 10-dB SNR, and (c) 3-dB SNR.

to as first window). The CC is instead evaluated only on the time window corresponding to the action potential duration (referred as second window, indicated for example in Fig. 6 as MUAP). The choice of the duration of the first window is not critical. The only limitation refers to the necessity of selecting a length longer than the time the MUAP's need to move along all the considered electrodes.

After recognizing the MUAP's with energy higher than a preset threshold in the first window and estimating their conduction velocity, the waveforms of these MUAP's are replaced by zeros in the reference EMG signal and the process continues until the whole reference signal is analyzed. The algorithm then restarts from the beginning of the modified signal until all MUAP's are detected. In each stage, the highest-energy MUAP's are detected and their CV values are estimated. The threshold for each first window is set as a percentage of the absolute maximum of the detection variable. This threshold is not critical and can be even set to 100%; in this case only the absolute maximum will be detected for each CWT calculation and for each MUAP detected one CWT calculation will be performed. If more than one maxima are detected for a CWT calculation, computational load is reduced (this is the only advantage in fixing the threshold lower than 100%). The process ends when either no MUAP's of acceptable duration are found or no value of CC in the beamforming algorithm is above the threshold. In addition to the estimation of conduction velocity the algorithm gives the extracted MUAP's and can be used as a starting point for MUAP classification and decomposition of the surface EMG signal [26].

The previously described procedures are implemented in MATLAB-5 language.

III. Results

The performance of the described technique has been evaluated a) using simulated EMG signals having *a priori* known conduction velocity distribution and corrupted with white noise, and b) using real EMG signals.

Fig. 5 shows a 300-ms-long segment from a 1-s-long simulated double differential EMG signal, constituted by 92

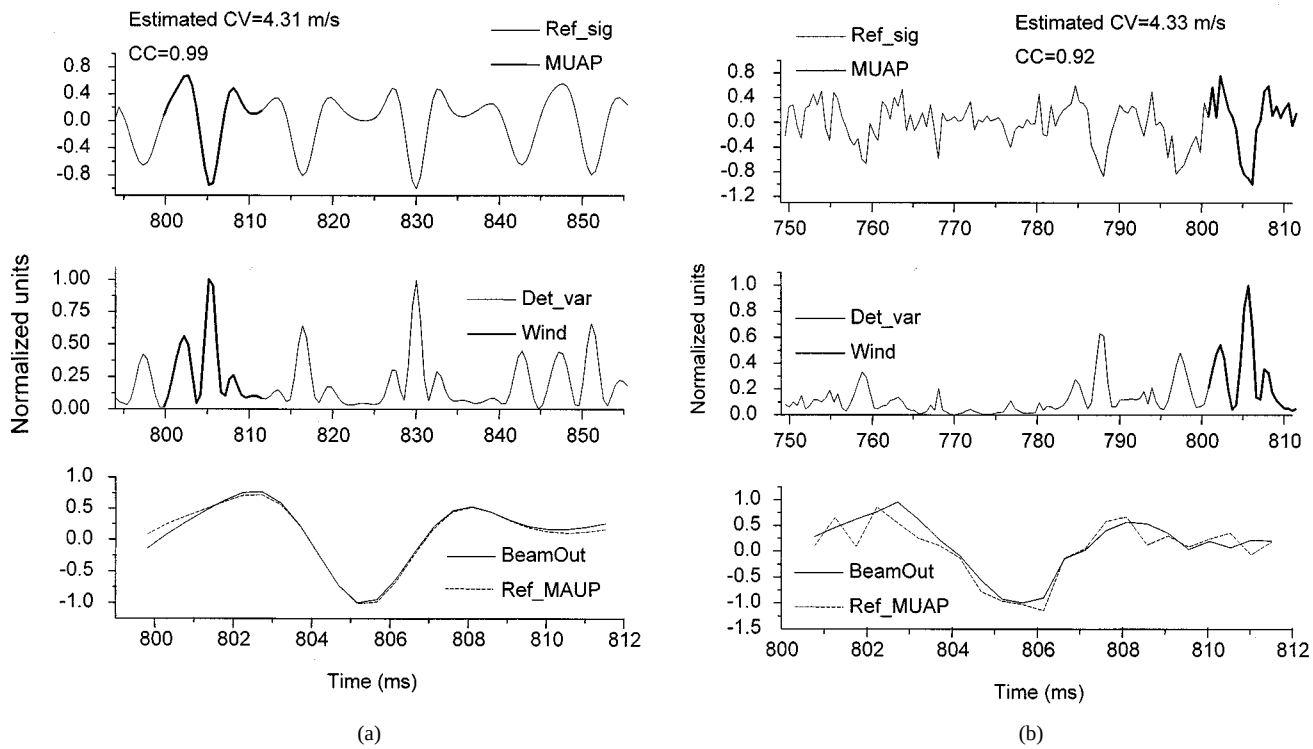

Fig. 6. Examples of MUAP identification for a simulated signal; the reference signal and the identified MUAP are reported in the top panel; the detection variable and the selected MUAP window (wind) are shown in the middle panel; the reference MUAP (- -) and the beamforming's output (-) are presented in the bottom panel. The preset CV was 4.30 m/s. The SNR is (a) ∞ and (b) 3 dB. The same MUAP (800–812 ms) is detected with and without noise.

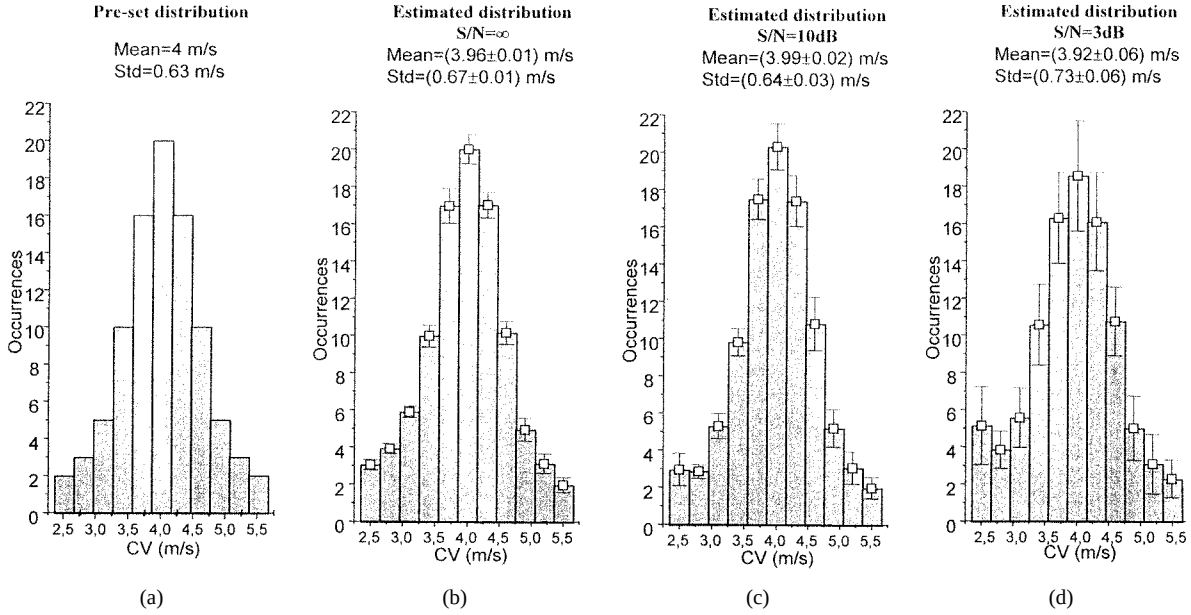

Fig. 7. (a) Preset CV distribution and (b) estimated CV distribution for the noise-free signal (c) for the noise corrupted signal with SNR of 10 dB, and (d) for the noise-corrupted signal with SNR of 3 dB. The mean and standard deviation of the distributions are shown as well as the mean and standard deviation of the histograms' bins.

MUAP's from 11 MU's with some cases of partially overlapping MUAP's (in the simulation seven channels of the array are generated), in the case of infinite SNR, of 3-dB and of 10-dB SNR (SNR is the same for all the channels of the array). 200 different signals of this type have been generated in the case of infinite, 10- and 3-dB SNR and the CV distribution has been computed with the described technique. The simulated signals have been generated with the same CV distribution but with MUAP's placed in random positions; the superposition was anyway maintained always below the complete overlapping. Fig. 6 presents the different steps of the detection and CV estimation technique. Even at high levels of added noise, the algorithm identifies each MUAP's and the error on the CV estimation is acceptable. Note that the noise is considerably

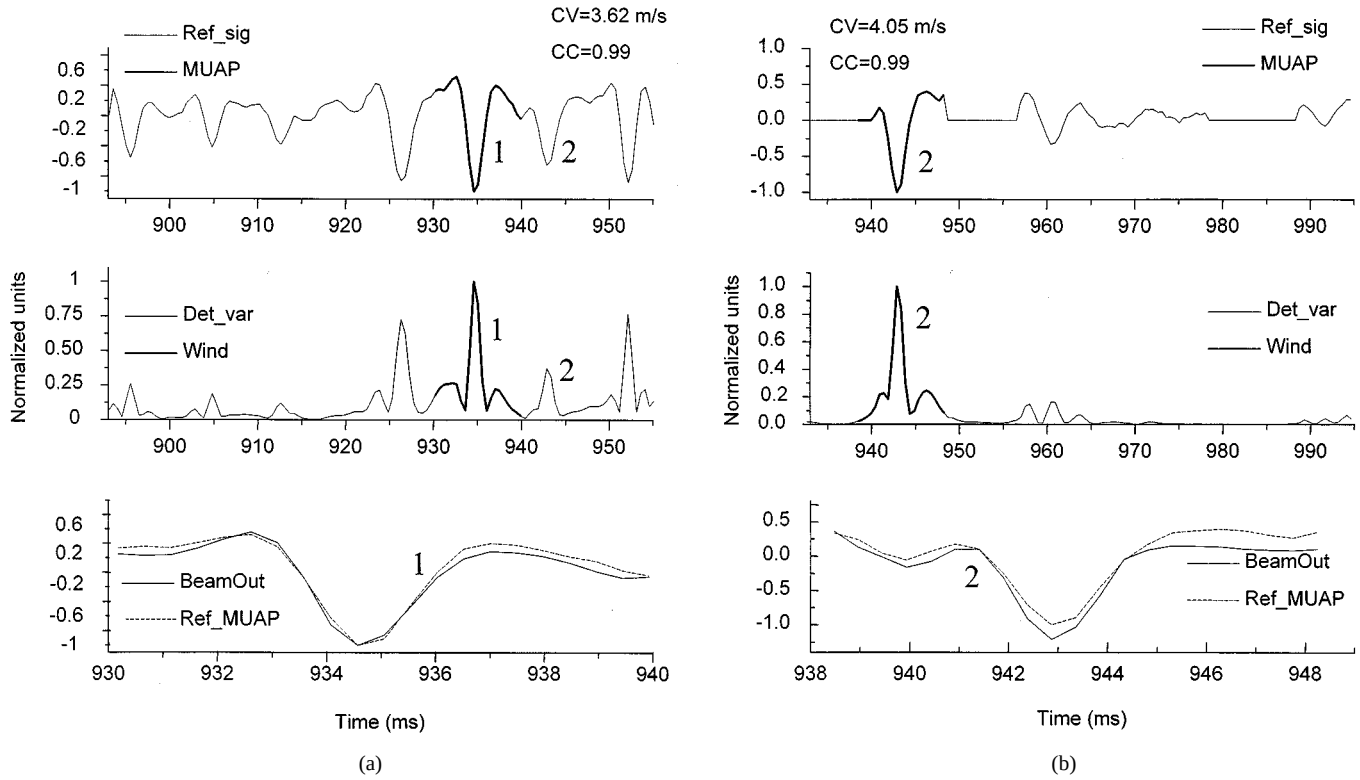

Fig. 8. Example of high energy 1) and low energy 2) MUAP identification [(a) and (b), respectively] in a real signal. The reference signal and the identified MUAP are reported in the top panel, the detection variable and the selected MUAP window (wind) are shown in the middle panel, the reference MUAP (- -) and the beamforming's output (-) are presented in the bottom panel.

attenuated in the beamforming output. The conduction velocity distributions estimated in different conditions of noise level are reported in Fig. 7 and compared to the preset distribution. Mean and standard deviation of the estimated distributions are also shown. The results are quite accurate even in the worst case, especially regarding the estimation of the mean and standard deviation of the distribution.

Several interesting considerations can be drawn by testing the technique with a 60-s-long signal, recorded from the right biceps brachii muscle in a normal subject during a 50% MVC contraction. The interelectrode distance of the 16-electrode array was 5 mm (electrode size $1\times$ 10 mm) and the sampling rate was 2048 Hz. Fig. 8 shows two different cases of MUAP identification. Fig. 8(a) presents the identification of the high-energy MUAP and Fig. 8(b) the identification of the subsequent in time low energy MUAP. Note that the identification of the second MUAP is performed on a signal in which most of the contributions have been deleted after the estimation of their CV. The correlation between the reference signal and the beamforming's output is very high in the two situations. Fig. 9 shows the evident change of CV distribution of the MUAP's during the contraction. The distribution shows the number of firings in a specific bin.

Fig. 10 depicts the results of the application of the technique in the case of a voluntary contraction of a healthy biceps at 60% MVC; in the figure the plot of the conduction velocity estimated with spectral matching algorithm [9], [10] is also reported. It is evident that the values obtained with the global method are in agreement with those given by the estimation of CV distribution.

IV. Discussion

Results have shown that the use of linear electrode arrays, together with good spatial selectivity and appropriate algorithms makes it possible to determine, from noninvasive measurements, the CV values of individual MUAP's with a resolution that has been matched in the past only by invasive procedures.

This technique has shown many favorable properties: 1) it is noninvasive and easy to apply, 2) it is quite reliable, and 3) in the presence of superimposed MUAP's, if the overlapping portions of waveforms are relatively limited, the technique is able to distinguish each MUAP and estimate their conduction velocity separately. When the degree of superimposition is high, the compound MUAP's are detected as a single propagating signal and a CV value which is a weighted average of the different values is estimated.

Some important limitations have to be underlined for a correct use of the method. As presented in Fig. 1 and explained in the text, the channels used for the estimation correspond to unidirectional propagation of the action potentials between the innervation zone and the tendon region. In the case of presence of more than one innervation zone the channels chosen must be between the innervation zone nearest to tendon and the tendon itself and their number may be small. The case of multiple innervation zones is not common in the biceps brachii but quite common in other muscles, such as the vastus lateralis and medialis. Furthermore the assumption of equal delays along the array

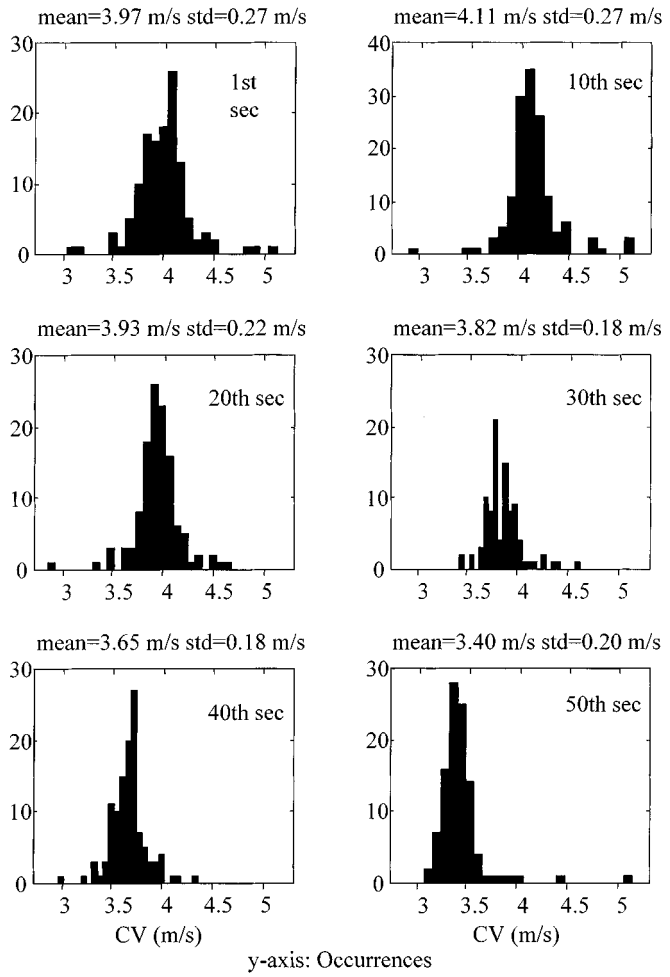

Fig. 9. Estimated CV distribution for a 50% MVC signal during a 60-s-long voluntary contraction of a biceps brachii; each histogram has been calculated during 1 s of contraction.

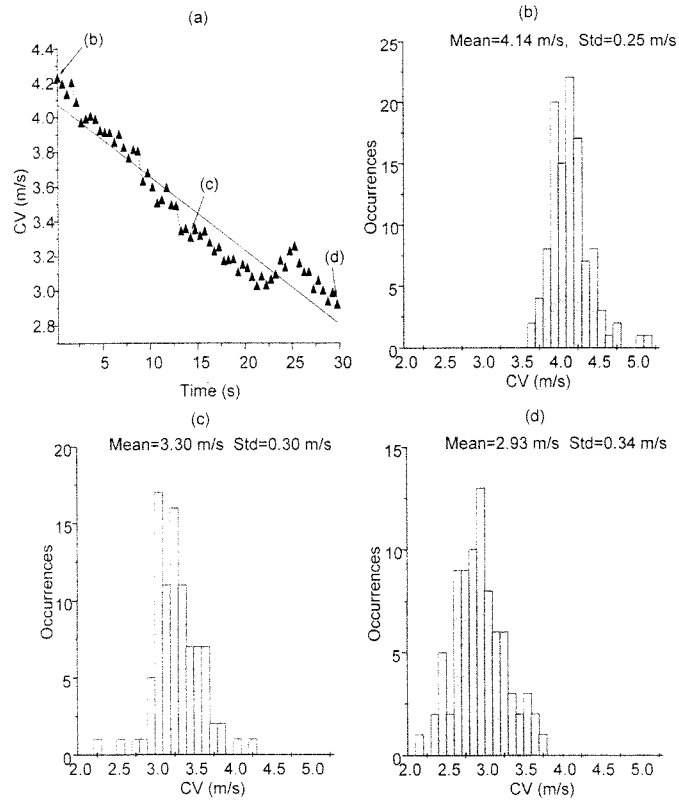

Fig. 10. CV estimated with spectral matching method (a) and estimations of CV distribution calculated for three signal epochs (b)–(d) during an isometric contraction sustained at 60% MVC for 30 s. Each histogram has been calculated during 1 s epochs while the global values (a) have been calculated during epochs of 0.5 s.

is correct only in the case of perfect alignment of the array with respect to the fibers [27], [28] and homogeneity of the tissues [29].

Moreover, this technique provides a histogram of the CV values of MUAP's, not of MU's. A MU firing at high frequency will increase the events in the bin corresponding to its CV value. This problem will be solved by classification of the MUAP's in groups belonging to the same MU [26].

Another issue to be discussed concerns the case of superimposed MUAP's which are sometimes detected as one single MUAP and for which one single conduction velocity value is obtained; the CV estimate obtained in this case is the weighted average (weighted by the energies of the contributions) of the CV's of the superimposed MUAP's. The problem is negligible in the case of small number of superpositions but becomes serious when the contraction level is high and high energy MUAP's mask the low energy ones. In the latter case the conduction velocity distribution estimation is polarized by the largest MUAP's. With the proposed approach this problem can not be solved but only attenuated by improving the spatial selectivity of the detection system [20], [21]. Moreover, some practical methods of measurements can be suggested; for example instead of applying the algorithm to signals detected at high contraction levels it would be preferable to apply it at different contraction levels, starting from very low levels, and estimate the conduction velocities first from signals detected at low levels, identifying the smallest MUAP's, and then from signal at high levels, identifying the largest MUAP's. A simple measurement protocol can be based on linearly varying force contractions; the algorithm can then be applied to the entire signal thereby obtaining information on the recruitment order.

Despite the outlined limitations the technique has many advantages with respect to the previously available methods [11] and is easily applicable to real cases, with potential clinical relevance [30].

V. Conclusions

The results show that the proposed technique can give reliable information about the CV distribution of the motor units, as evaluated with simulation studies. The described technique to establish CV distribution can be easily implemented in clinical EMG laboratories to study motor unit discharge patterns and recruitment order in patients with neuromuscular disorders, it is easy to apply and, with the limitations described above, quite accurate. Preliminary results concerning clinical application are described in [30].

APPENDIX
PROPERTIES OF THE HR FUNCTIONS USED AS MATCHED WAVELETS

The HR functions are defined as

$$w_{\lambda,n}(t) = \frac{1}{\sqrt{2^n n!}} H_n\left(\frac{t}{\lambda}\right) \frac{1}{\sqrt{\pi}\lambda} e^{-t^2/\lambda^2} \quad n \in [0, \infty) \quad (5)$$

where

n is the order of the function

λ is a width parameter

$H_n(t)$ are the Hermite polynomials recursively defined as

$$\begin{cases} H_0(t) = 1 \\ H_1(t) = 2t \\ H_n(t) = 2tH_{n-1}(t) - 2(n-1)H_{n-2}(t). \end{cases} \quad (6)$$

The HR function of order n is proportional to the nth derivative of a Gaussian function [24], [31]. Its expression and Fourier transform are

$$w_{\lambda,n}(t) = \frac{(-1)^n \lambda^n}{\sqrt{2^n n!}} \frac{d^n}{dt^n}\left(\frac{1}{\sqrt{\pi}\lambda} e^{-t^2/\lambda^2}\right) \quad (7)$$

$$\Im[w_{\lambda,n}(t)] = \frac{(-j)^n}{\sqrt{2^n n!}} (2\pi f \lambda)^n e^{-(2\pi f \lambda/2)^2}. \quad (8)$$

So, for any λ and $n = 2$ (HR functions of order two), the Fourier transform of the HR function is an even function negative for all f. The HR function of order two has been chosen as mother wavelet in this study. The issue is to demonstrate that the maximum point of the scalogram in time is the same for all constant scale sections when the mother wavelet is matched with the signal. Let's consider the CWT of the function $f(t)$

$$WT(a, \tau) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \cdot h^*\left(\frac{t-\tau}{a}\right) dt \quad (9)$$

and let us suppose that the wavelet is real and, scaled of a factor a_0, matches perfectly with the real function $f(t)$ (in the case of the HR functions this means that the λ value that gives the perfect matching corresponds to a_0 in the CWT). Without loss of generality, we may assume $a_0 = 1$

$$h(t) = h^*(t) = f(t). \quad (10)$$

The scalogram is defined as the square of the absolute value of the CWT

$$|WT(a, \tau)|^2 = \left|\frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \cdot h\left(\frac{t-\tau}{a}\right) dt\right|^2. \quad (11)$$

Considering the scale $a = 1$ it can be shown that the maximum point corresponds to $\tau = 0$, as a consequence of the matched filter theory. The same property can be demonstrated for any scale a by using Parseval's theorem [32] and the hypothesis (10)

$$\begin{aligned} |WT(a, \tau)|^2 &= \left|\frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(t) \cdot f^*\left(\frac{t-\tau}{a}\right) dt\right|^2 \\ &= \left|\frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} F(f) \cdot a \cdot F(af) \cdot e^{-j2\pi f\tau} df\right|^2 \end{aligned} \quad (12)$$

with $F(f)$ being the Fourier transform of $f(t)$.

Using a well-known property of the absolute value of an integral, it can be written

$$\begin{aligned} \gamma(a, \tau) &= \left|\frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} F(f) \cdot a \cdot F(af) \cdot e^{-j2\pi f\tau} df\right| \\ &\le \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} |F(f) \cdot a \cdot F(af) \cdot e^{-j2\pi f\tau}| df \\ &= \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} |F(f) \cdot a \cdot F(af)| df. \end{aligned} \quad (13)$$

In the case of function with even or odd symmetry, the Fourier transform of $f(t)$ has null imaginary or real part, respectively, [32] and the argument of the integral is always real. Furthermore, if the argument is also positive (i.e., if the Fourier transform of the function, or its imaginary part in the case of odd symmetry of the function $f(t)$, is either positive or negative or positive for $f > 0$ and negative for $f < 0$ or vice versa), the operation of absolute value and integral commute (assuming $a > 0$) and it can be written

$$\begin{aligned} \gamma(a, \tau) &\le \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} |F(f) \cdot a \cdot F(af)| df \\ &= \left|\frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} F(f) \cdot a \cdot F(af) df\right| \\ &= \gamma(a, 0). \end{aligned} \quad (14)$$

The second member of (14) is the absolute value of the CWT evaluated in $\tau = 0$, which is, then, the absolute maximum of the scalogram for any scales

$$|\gamma(a, \tau)|^2 = |WT(a, \tau)|^2 \le |\gamma(a, 0)|^2 = |WT(a, 0)|^2. \quad (15)$$

From (8) it appears that the HR functions of any order n satisfy the desired conditions; in fact their Fourier transform (or its imaginary part) is even (for n even) or odd (for n odd) and does not change sign for positive or negative frequencies.

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